

Chris Dai

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Education

Harvard University

M.S. Health Data Science, T.H. Chan School of Public Health
University of Michigan, Ann Arbor

B.S.E. Industrial and Operations Engineering; Minor in Computer Science
UM-SJTU Joint Institute, Shanghai Jiao Tong University

B.S.E. Electrical and Computer Engineering

Dean's Honor List (2024, 2025) · Myun Lee Scholarship (2025)

Aug. 2026 – Expected May 2028

Boston, Massachusetts, USA

Aug. 2024 – May 2026

Ann Arbor, Michigan, USA

Sep. 2022 – Aug. 2026

Shanghai, China

Skills

LLM & Evaluation: SFT corpus construction, synthetic dialogue generation & data flywheels, LLM-as-judge evaluation, RAG & embedding retrieval (bge-m3), prompt architecture, agent benchmarking

ML & Data: PyTorch, K-means / PCA, time-series feature engineering, SQL, PostgreSQL, dbt, evaluation-harness design

Engineering: Python, TypeScript, JavaScript, FastAPI, Flask, React, Node.js, Next.js, Playwright / Chrome DevTools Protocol

Experience

AI Speech

Jun. 2026 – Aug. 2026

Sponsored Capstone Thesis — LLM data flywheel for in-vehicle voice assistants

Shanghai, China

- Led the data flywheel end to end for the multimedia domain of a 5-person voice-assistant project: built a neuro-symbolic user simulator — **41** persona archetypes over an agenda state machine — emitting retrieval-grounded rollouts against a bge-m3 tool index.
- Reward-scored every rollout on the team's 397B model on a GPU cluster, keeping only provably correct (reward=1) traces, then regenerated 200 dialogues for each of the 11 highest-error functions; spurious no-tool calls fell **4,424**→**1,410**.
- Produced **17,200** verified (reward=1) multimedia fine-tuning dialogues — 15,000 base plus 2,200 error-targeted — as a four-quadrant corpus (positive/negative × single/multi-turn), each batch cleared by the same automated quality gate.
- Fine-tuned the assistant's model on this corpus, lifting tool-call accuracy **72.8%**→**82.7%** on real conversations, **76.8%**→**98.3%** on the synthetic validated set and **83.4%**→**96.0%** on the user-profile set — graded by an independent DeepSeek-V4-Pro judge, not the 397B corpus scorer.

ByteDance

May 2026 – Aug. 2026

AI Solutions Intern; Stack: Python, Flask, React, TypeScript, Node.js, Playwright / CDP

Shanghai, China

- Built an **LLM-agent evaluation platform** for two enterprise desktop agents: designed a suite of **200+** real user tasks, drove both Electron clients over the Chrome DevTools Protocol, normalized their incompatible telemetry into one trace schema, and scored them with an LLM-as-judge harness on output quality, token throughput and cost — one comparable read, and the team's basis for selection.
- Productized customer technical due diligence into a **citation-grounded RAG** research tool on internal AI tooling, where every conclusion resolves to its source; adopted by the regional client team as their working tool.
- Distilled the team's enterprise deep-research workflow into reusable team-wide skills: one research pass, **2 h** → **5 min**.
- Hand-coded all **11** agents' skills for a client-facing demo site — industry-specific and general-purpose — including a PPTX-to-HTML skill the team still uses daily, plus **13** case studies on a shared asset base of **240+** industry reference materials.

Shenzhen Legion Technology Co., Ltd.

Apr. 2025 – Apr. 2026

Data Engineering & AI Intern (part-time); Stack: Python, FastAPI, PostgreSQL, React/TS

Remote, China

- Co-developed **MetaFlow** (two-person project, started Mar. 2026), a metadata-centric NL-to-SQL system that answers questions through a verifiable relational IR instead of generating SQL directly; I owned the metadata layer, evaluation suite and studio front end — my collaborator owned the query engine.
- Built its gold-case regression suite — **20** scenarios pinned across 5 pipeline stages as contract-as-test — plus a contracts endpoint running data-quality rules as executable checks, so a schema change is caught by re-running the suite, not by hand.
- Earlier in the role, designed the prompt architecture of an open-source RAG data-governance copilot for dbt pipelines — question classification (catalog / SQL / data-QA), entity extraction, embedding-based semantic retrieval — and its evaluation framework of fixed test cases for retrieval accuracy and governance compliance, owning feature scope and release priorities, not the upstream codebase.

PGF Technology Group

Mar. 2026 – Apr. 2026

Consulting Engagement (4-person team), sole developer; Stack: Next.js, TypeScript, Postgres

Rochester Hills, Michigan, USA

- Sole developer of the software on a four-person consulting engagement: shipped a full-stack work-order tracking system covering all **9** production stations of a ~50-person harness manufacturer — a complete solution the client took into internal use after handover.
- Diagnosed stale reads across serverless instances and migrated storage in production (in-memory → Blob → Postgres); pinned the scan, queue and alert flows with **6** Playwright e2e suites.

Projects

Cardiac Rehabilitation Wearable-Data Study

Apr. 2025 – Jan. 2026

SURE Research Program, University of Michigan — Advisors: Profs. Mariel Lavieri & Jessica Golbus

Ann Arbor, Michigan, USA

- Built a time-series feature pipeline over **400,000+** raw step records from **247** enrolled cardiac-rehab patients — rolling means, trend slopes, periodicity, plus imputation for gapped and high-variance series.
- Phenotyped recovery trajectories with K-means/PCA and detected plateaus with a Kalman filter: against the standard 36-session protocol, ~**68%** of the 191-patient analysis cohort was discharged off its true plateau.

EMG-Driven Rehab Training Game & Muscle-Synergy Analysis

Oct. 2024 – Apr. 2026

NeuRro Lab, University of Michigan — PI: Chandramouli Krishnan

Ann Arbor, Michigan, USA

- Built the lab's EMG-driven rehabilitation training game **end to end** — Unity/C# game client and in-game UI, real-time signal pipeline, and the backend database — now a standard tool in ongoing clinical experiments.
- Engineered its real-time data path — **16-channel** EMG at **2 kHz** through band-pass, envelope extraction, EMA smoothing and MVC calibration into cursor control — and ported its multi-objective gradient-descent optimizer from **PyTorch** to C# for on-device adaptation across an **11-stage** protocol.
- Built the lab's muscle-synergy pipeline in MATLAB — NNMF-NNR, cross-subject alignment, group statistics over **18** subjects; in clinical use.